THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 0.68 m	v/f.
2	**(b)** greater for the 256 Hz fork	lambda proportional to 1/f.
3	**(d)** quality or timbre	Waveform.
4	**(a)** 20	3.4/0.17.
5	**(c)** 0.00588 s	lambda/v.
6	**(c)** increases	Larger amplitude.
7	**(d)** half a wavelength	Half cycle.
8	**(b)** decibels	dB.
9	**(c)** 343 m/s	f lambda.
10	**(c)** becomes higher	Higher frequency.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	v = 340 m/s (1); T = 0.05 s; 10 oscillations 0.5 s (1).	1+1
12	Sound travels faster in helium (1); Frequency of resonance increases, so pitch is higher (1).	1+1
13	Measure the time interval between flash and bang at a known distance (1); v = distance/time (1).	1+1
14	Distance 102 m (1); t = 0.3 s (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	f = 400 Hz (1); T = 0.0025 s (1); lambda = 0.85 m (1).	1+1+1
16	t = d/v (1); 3400/340 (1); 10 s (1).	1+1+1
17	Megaphone: conical tube directs sound (1); Hearing aid amplifies sound (1); Converts to electric signals and back (1).	1+1+1
	OR: Depth = 1500 m (1); Double depth: 3000 m (1); Time = 4 s (1).	1+1+1
18	lambda = 340/50000 = 6.8 mm (1); Time = 3.4/340 (1); = 0.01 s (1).	1+1+1
	OR: Amplitude of source (1); Distance from the source (1); Sensitivity of the ear (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Nearer cliff = 170 m (1); Farther cliff = 340 m (1); Distance between cliffs = 510 m (1); At the centre: 255 m each (1); Time = 510/340 = 1.5 s for both echoes (1).	1+1+1+1+1
	OR: A is taller (1); B has more waves in the same time (1); A louder (1); B higher pitch (1); Wavelengths: B is half of A (1).	1+1+1+1+1
20	Pinna collects sound (1); Auditory canal and eardrum (1); Ossicles amplify (1); Cochlea and nerve to the brain (1); Precautions: avoid loud noise, no sharp objects (1).	1+1+1+1+1
	OR: lambda20 = 344/512 = 0.672 m (1); lambda40 = 356/512 = 0.695 m (1); Frequency is the same 512 Hz (1); Wavelength increases with temperature (1); Speed increases (1).	1+1+1+1+1

Mistakes to avoid:

- Forgetting to halve the total distance in echo and SONAR problems: the sound goes to the obstacle *and back*, so $d = vt/2$.
- Mixing up pitch and loudness: pitch depends on frequency, loudness on amplitude.
- Drawing sound as a transverse wave or saying that the particles travel with the wave; in air the particles vibrate to and fro along the direction of travel.
- Saying that sound travels fastest in air and slowest in solids; the order is solids $>$ liquids $>$ gases.
- Writing the audible range as 20 kHz to 20 Hz, or calling a sound of 25 kHz infrasonic.

- Applying $v = f\lambda$ with wavelength in cm or frequency in kHz without converting to m and Hz, or saying that the speed of a wave changes when only the frequency is changed in the same medium.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** time period	T.
2	**(a)** less loud	Lower level.
3	**(d)** 100	f t.
4	**(b)** high pitch	Thin voices.
5	**(a)** 400 Hz	N/t.
6	**(d)** 0 dB	Faintest.
7	**(c)** 255 m	vt/2.
8	**(b)** lower frequency	Deep.
9	**(c)** an infrasonic wave	Below 20 Hz.
10	**(d)** pain and hearing damage	Dangerous.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Different waveform (quality) (1); Overtones differ (1).	1+1
12	Vibrations travel along the tight string (1); Solid medium transmits sound (1).	1+1
13	d = v t/2 (1); = 344 x 0.1/2 = 17.2 m (1).	1+1
14	f = v/lambda (1); = 1000 Hz (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	d = 340 m (1); Time = 680/350 (1); = 1.94 s (1).	1+1+1
16	Solids are more elastic (1); Particles are closely packed (1); Disturbance passes quickly (1).	1+1+1
17	Reflections from walls add (1); Sound cannot spread out (1); Reverberation (1).	1+1+1
	OR: lambda = 0.6 m (1); Half of lambda = 0.3 m (1); v = f lambda (1).	1+1+1
18	Cooler air near the ground at night (1); Bends sound back towards the ground (1); Hence sound is heard farther (1).	1+1+1
	OR: Ultrasonic pulses sent through metal (1); Cracks reflect them (1); Echo shows the flaw (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Distance = 1500 x 0.8/2 = 600 m (1); lambda = 1500/40000 = 0.0375 m (1); Waves in 0.01 s = 400 (1); Ultrasound is directional (1); Short wavelength gives sharper detection (1).	1+1+1+1+1
	OR: Pulse and echo (1); Depth = vt/2 (1); = 1500 x 0.04/2 (1); = 30 m (1); Stronger echo from the sea bed (1).	1+1+1+1+1
20	Stand at a measured distance from a wall (1); Clap and time the echoes (10 repeats) (1); Average time t (1); v = 2d/t (1); e.g. 10 echoes in 5 s: t = 0.5 s, v = 340 m/s (1).	1+1+1+1+1
	OR: Bell in a jar (1); Pump out air (1); Sound fades (1); Radio waves carry the messages (1); Radio needs no medium (1).	1+1+1+1+1

Mistakes to avoid:

- Forgetting to halve the total distance in echo and SONAR problems: the sound goes to the obstacle *and back*, so $d = vt/2$.
- Mixing up pitch and loudness: pitch depends on frequency, loudness on amplitude.
- Drawing sound as a transverse wave or saying that the particles travel with the wave; in air the particles vibrate to and fro along the direction of travel.
- Saying that sound travels fastest in air and slowest in solids; the order is solids $>$ liquids $>$ gases.
- Writing the audible range as 20 kHz to 20 Hz, or calling a sound of 25 kHz infrasonic.
- Applying $v = f\lambda$ with wavelength in cm or frequency in kHz without converting to m and Hz, or saying that the speed of a wave changes when only the frequency is changed in the same medium.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 68.8 m	344 x 0.4/2.
2	**(c)** higher frequency	Frequency decides pitch.
3	**(a)** ultrasonic	Above 20 kHz.
4	**(c)** their high frequency vibrations dislodge dirt	Vibration.
5	**(a)** 1800 m	vt/2.
6	**(c)** frequency	Source decides frequency.
7	**(d)** audible to dogs but not to humans	Ultrasonic for humans.
8	**(d)** second	T is measured in seconds.
9	**(b)** 5 s	1700/340.
10	**(c)** 1.5 m	v/f.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sound travels fast and far in water (1); Use low-frequency sounds that travel far (1).	1+1
12	Particles of warm air have more energy and move faster (1); Disturbance is passed on quickly (1).	1+1
13	15 Hz is below the audible range (1); 15 kHz lies within 20 Hz to 20 kHz (1).	1+1
14	Amplitude increases (1); Loudness increases (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Angle of incidence equals angle of reflection (1); Megaphone directs sound in one direction (1); Sound boards spread sound over a hall (1).	1+1+1
16	26400 vibrations (1); T = 0.00227 s (1); f = N/t (1).	1+1+1
17	lambda max = 340/20 = 17 m (1); lambda min = 340/20000 = 0.017 m (1); Range 1.7 cm to 17 m (1).	1+1+1
	OR: Pulses reflect from tissues (1); Echoes give images (1); No harmful ionising radiation (1).	1+1+1
18	Distance in water 750 m (1); Distance in steel 2500 m (1); Ratio 1 : 3.3 (1).	1+1+1
	OR: Frequency remains the same (1); Wavelength increases as speed increases (1); Amplitude depends on the energy (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Air: lambda = 0.2 m, 170 waves (1); Time = 0.1 s (1); Medium: lambda = 1 m (1); 34 waves in 34 m, time 0.02 s (1); Frequency unchanged (1).	1+1+1+1+1
	OR: Medium (1); Type of wave (1); Speed (1); Frequency range (1); Both show reflection and refraction (1).	1+1+1+1+1
20	Megaphone (1); Stethoscope (1); Sound boards (1); Concert halls curved ceilings (1); Absorbing materials (1).	1+1+1+1+1
	OR: Frequency remains 440 Hz (1); lambda air = 0.77 m (1); lambda water = 3.41 m (1); lambda steel = 11.4 m (1); Speed and wavelength increase together (1).	1+1+1+1+1

Mistakes to avoid:

- Forgetting to halve the total distance in echo and SONAR problems: the sound goes to the obstacle *and back*, so $d = vt/2$.
- Mixing up pitch and loudness: pitch depends on frequency, loudness on amplitude.
- Drawing sound as a transverse wave or saying that the particles travel with the wave; in air the particles vibrate to and fro along the direction of travel.
- Saying that sound travels fastest in air and slowest in solids; the order is solids $>$ liquids $>$ gases.
- Writing the audible range as 20 kHz to 20 Hz, or calling a sound of 25 kHz infrasonic.

- Applying $v = f\lambda$ with wavelength in cm or frequency in kHz without converting to m and Hz, or saying that the speed of a wave changes when only the frequency is changed in the same medium.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND